

Let E be an elementary matrix ($n \times n$). Then
 $\det(E) = \det(E^T)$

Proof

① Let E be the matrix that performs the row operation
 $R_i \rightarrow R_i + \alpha R_j$

Then $E = I_n + \alpha E_{ij}$. Therefore $E^T = (I_n + \alpha E_{ij})^T$
 and $\det(E) = 1$ (See earlier lectures) $= I_n^T + \alpha E_{ij}^T$
 $= I_n + \alpha E_{ji}$

So, E^T is the matrix that performs the row operation

$$R_j \rightarrow R_j + \alpha R_i$$

Thus $\det(E^T) = 1$

Hence $\det(E) = 1 = \det(E^T)$.

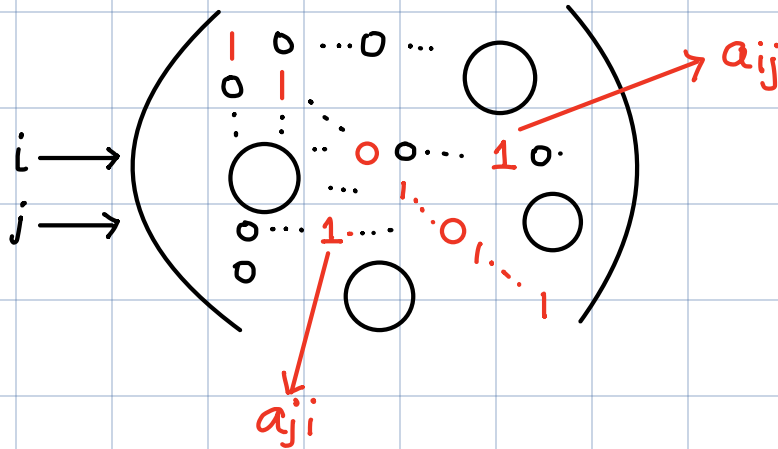
② If E is the matrix obtained by performing the row operation
 $R_i \rightarrow \alpha R_i$ then

$$E = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \alpha \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \text{ and therefore } E = E^T.$$

Therefore $\det(E) = \det(E^T)$.

© If E is the matrix obtained by switching rows $R_i \leftrightarrow R_j$ then

$$E = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji} \quad i < j$$



Then

$$\Rightarrow E^T = I_n - E_{ii} - E_{jj} + E_{ji} + E_{ij} = E.$$

Therefore $\det(E^T) = \det(E)$

④ Let A be an $n \times n$ matrix. Then

$$\det(A) = \det(A^T)$$

Proof

Case (i) Let A be invertible.

Then, we know that $A = E_1 E_2 \cdots E_n$, where each E_i is an elementary matrix.

Now $A^T = E_n^T E_{n-1}^T \cdots E_2^T E_1^T$ (why? how?)

and therefore,

$$\begin{aligned} \det(A^T) &= \det(E_n^T) \cdots \det(E_1^T) \\ &= \det(E_n) \cdots \det(E_1) \\ &= \det(E_1) \cdots \det(E_n) \\ &= \det(E_1 E_2 \cdots E_n) \\ &= \det(A). \end{aligned}$$

Case (ii) Let A be noninvertible matrix

Then,

$E_1 E_2 \cdots E_n A = J$, where J is the row-reduced echelon form of A . Then the n^{th} row of J is a zero row.

We have, $A^T E_n^T \cdots E_1^T = J^T$

$$\Rightarrow \det(A^T) \det(E_n^T) \cdots \det(E_1^T) = \det(J^T).$$

Since the n^{th} column of J^T is a zero column, we must have $\det(J^T) = 0$ (why? see Tutorial 5).

Now since $\det(E_i^T) = \det(E_i)$ and $\det(E_i) \neq 0$, we must have $\det(A^T) = 0$.

Thus in this case $\det(A) = 0 = \det(A^T)$ □

Calculating inverse of a matrix

Formula for determinant

 (Expanding determinant via any row)

Let A be an $n \times n$ matrix and A_{ij} be the matrix obtained by removing the i^{th} row and j^{th} column of A .

Then, for any k , $1 \leq k \leq n$,

$$\det(A) = \sum_{l=1}^n (-1)^{k+l} a_{kl} \det(A_{kl})$$

Proof: Let B be the matrix obtained from A through the following row operations

$$R_k \leftrightarrow R_{k-1}, R_{k-1} \leftrightarrow R_{k-2}, \dots, R_2 \leftrightarrow R_1.$$

For Example, if $n=4$ and $k=3$ and

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} \text{ then } B = \begin{pmatrix} a_{31} & a_{32} & a_{33} & a_{34} \\ a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Then it is easily seen that for any l , $1 \leq l \leq n$, the matrix obtained by deleting the 1^{st} row and l^{th} column of B

is the same as the matrix obtained by deleting the k^{th} row and l^{th} column of A . [That is, $B_{l,l} = A_{k,l}$]

Note that $\det(A) = (-1)^{k-1} \det(B)$.

$$\begin{aligned} \text{Thus } \det(B) &= \overset{b_{11}}{\overset{11}} a_{k1} \det(B_{11}) - \overset{b_{12}}{\overset{11}} a_{k2} \det(B_{12}) + \cdots + (-1)^{n-1} \overset{b_{1n}}{\overset{11}} a_{kn} \det(B_{1n}) \\ &= a_{k1} \det(A_{k1}) - a_{k2} \det(A_{k2}) + \cdots + (-1)^{n-1} a_{kn} \det(A_{kn}) \end{aligned}$$

and

$$\begin{aligned} \det(A) &= (-1)^{k-1} a_{k1} \det(A_{k1}) + (-1)^k a_{k2} \det(A_{k2}) + \cdots + (-1)^{n+k-2} a_{kn} \det(A_{kn}) \\ &= (-1)^{k+1} a_{k1} \det(A_{k1}) + (-1)^{k+2} a_{k2} \det(A_{k2}) + \cdots + (-1)^{k+n} a_{kn} \det(A_{kn}) \\ &= \sum_{l=1}^n (-1)^{k+l} a_{kl} \det(A_{kl}) \end{aligned}$$

□